Mo Zhou

Contact

10450 NE 10th St
Bellevue, WA 98004
United States

Tel: (+1) ***********
Email: cdluminate@gmail.com
Website: cdluminate.github.io

Interests

- Deep Learning, Computer Vision, and Multimodal Models
- Artificial Intelligence Security, Robustness and Trustworthiness

Experience

- Amazon.com Services LLC, Ring AI — Bellevue, WA 98004
  Applied Scientist (Vision Language Model) — 01/2026 - Current
  Worked-on: Video Understanding, VLM Evaluation Metric
- Google Research, Computational Imaging Team — Mountain View, CA 94043
  Student Researcher (Computer Vision) — 05/2024 - 10/2025
  Worked-on: Image Restoration [C02], Face Restoration [J01], and Infinite Zoom
  Mentor: Hossein Talebi, Keren Ye, Mauricio Delbracio, Peyman Milanfar
- Microsoft Research, Applied Sciences Group — Redmond, WA 98052
  Research Intern (Deep Learning) — 05/2023 - 08/2023
  Worked-on: Image Generation + LLM
  Mentor: Kazuhito Koishida, Saeed Amizadeh
- Wormpex AI Research LLC — Bellevue, WA 98004
  Research Intern (Computer Vision) — 05/2022 - 08/2022
  Mentor: Haoxiang Li, Yiding Yang, Gang Hua
  Worked-on: Re-biasable Object Detection [J02]
- Xi'an Jiaotong University — Xi'an, Shaanxi 710049
  Institute of Artificial Intelligence and Robotics (IAIR)
  Research Assistant (Computer Vision) — 07/2020 - 06/2021
  Worked-on: Adversarial Order Attack [C07], Adversarial Ranking Defense [J04]
  Supervisor: Le Wang, Sanping Zhou

Education

- Johns Hopkins University — Baltimore, MD, USA 21218
  Dept. Electrical and Computer Engineering, Whiting School of Engineering
  Ph.D. Electrical and Electronics Engineering — 08/2021 - 12/2025
  Advisor: Vishal M. Patel
- Xidian University — Xi'an, Shaanxi, China 710071
  M.Eng. Pattern Recognition and Intelligent Systems. July, 2020 — 09/2017 - 06/2020
  Thesis: Coherent Visual-Semantic Embedding for Cross-Modal Retrieval
  Advisor: Zhenxing Niu
- Xidian University — Xi'an, Shaanxi, China 710126
  B.Eng. Electromagnetic Field and Wireless Technology. July, 2017 — 09/2013 - 07/2017
  Advisor: Zhenxing Niu

Publications

Google Scholar Profile: scholar.google.com/citations?user=BVIO95UAAAAJ
(Jul. 07 2026) Citations: 2026 H-Index: 11 i10-Index: 11
Other Identifiers: [ORCiD] [Semantic Scholar] [Web of Science] [DBLP]

JOURNAL ARTICLES: (2 TPAMI, 2 TMLR, 1 TMM)

[OpenReview] [arXiv] [J01] Mo Zhou, Keren Ye, Viraj Shah, Kangfu Mei, Mauricio Delbracio, Peyman Milanfar, Vishal M. Patel, Hossein Talebi, *"Reference-Guided Identity Preserving Face Restoration,"* Transactions on Machine Learning Research (TMLR), 2026.

[PDF] [arXiv] [J02] Mo Zhou, Yiding Yang, Haoxiang Li, Vishal M. Patel, Gang Hua, *"Deployment Prior Injection for Run-time Re-biasable Object Detection,"* IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. DOI: 10.1109/TPAMI.2026.3667914

[OpenReview] [arXiv] [J03] Yatong Bai, Mo Zhou, Vishal M. Patel, Somayeh Sojoudi, *"MixedNUTS: Training-Free Accuracy-Robustness Balance via Nonlinearly Mixed Classifiers,"* Transactions on Machine Learning Research (TMLR), 2024.

[PDF] [arXiv] [Github] [J04] Mo Zhou, Le Wang, Zhenxing Niu, Qilin Zhang, Nanning Zheng, Gang Hua, *"Adversarial Attack and Defense in Deep Ranking,"* IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2024. DOI: 10.1109/TPAMI.2024.3365699

[PDF] [J05] Le Wang, Mo Zhou, Zhenxing Niu, Qilin Zhang, Nanning Zheng, *"Adaptive Ladder Loss for Learning Coherent Visual-Semantic Embedding,"* IEEE Transactions on Multimedia (TMM), 2021. DOI: 10.1109/TMM.2021.3139210

CONFERENCE PAPERS: (3 CVPR, 3 ICCV, 1 ECCV, 1 NeurIPS, 1 ICLR, 1 AAAI)

[PDF] [C01] Mucong Ding, Bang An, Tahseen Rabbani, Chenghao Deng, Anirudh Satheesh, Souradip Chakraborty, Mehrdad Saberi, Yuxin Wen, Kyle Rui Sang, Aakriti Agrawal, Xuandong Zhao, Mo Zhou, Mary-Anne Hartley, Lei Li, Yu-Xiang Wang, Vishal M. Patel, Soheil Feizi, Tom Goldstein, Furong Huang, *"A Technical Report on "Erasing the Invisible": The 2024 NeurIPS Competition on Stress Testing Image Watermarks"* in NeurIPS 2025 Datasets and Benchmarks Track, 2025.

[PDF] [arXiv] [C02] Mo Zhou, Keren Ye, Mauricio Delbracio, Peyman Milanfar, Vishal M. Patel, Hossein Talebi, *"UniRes: Universal Image Restoration for Complex Degradations,"* in Proc. IEEE International Conf. on Computer Vision (ICCV), 2025.

[PDF] [arXiv] [C03] Kangfu Mei, Mo Zhou, Vishal M. Patel, *"Field-DiT: Diffusion Transformer on Unified Video, 3D, and Game Field Generation,"* in Proc. International Conference on Learning Representations (ICLR), 2025.

[PDF] [arXiv] [Github] [C04] Mo Zhou, Vishal M. Patel, *"On Trace of PGD-Like Adversarial Attacks,"* in Proc. International Conference on Pattern Recognition (ICPR), 2024.

[PDF] [Github] [C05] Yiqun Mei, Pengfei Guo, Mo Zhou, Vishal M. Patel, *"Resource-Adaptive Federated Learning with All-In-One Neural Composition,"* Advances in Neural Information Processing Systems (NeurIPS), 2022.

[PDF] [arXiv] [Github] [C06] Mo Zhou, Vishal M. Patel, *"Enhancing Adversarial Robustness for Deep Metric Learning,"* in Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2022.

[PDF] [arXiv] [Github] [C07] Mo Zhou, Le Wang, Zhenxing Niu, Qilin Zhang, Yinghui Xu, Nanning Zheng, Gang Hua, *"Practical Order Attack in Deep Ranking,"* in Proc. IEEE International Conf. on Computer Vision (ICCV), 2021.

[PDF] [arXiv] [Github] [C08] Liushuai Shi, Le Wang, Chengjiang Long, Sanping Zhou, Mo Zhou, Zhenxing Niu, Gang Hua, *"SGCN: Sparse Graph Convolution for Pedestrian Trajectory Prediction",* In Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2021.

[PDF] [arXiv] [Github] [C09] Mo Zhou, Zhenxing Niu, Le Wang, Qilin Zhang, Gang Hua, *"Adversarial Ranking Attack and Defense,"* in Proc. European Conf. on Computer Vision (ECCV), 2020.

[PDF] [arXiv] [Github] [C10] Mo Zhou, Zhenxing Niu, Le Wang, Zhanning Gao, Qilin Zhang, Gang Hua, *"Ladder Loss for Coherent Visual-Semantic Embedding,"* in Proc. AAAI Conf. on Artificial Intelligence (AAAI), 2020.

[PDF] [C11] Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, Gang Hua, *"Hierarchical Multimodal LSTM for Dense Visual-Semantic Embedding,"* in Proc. IEEE International Conf. on

Computer Vision (ICCV), 2017.

[PDF] [Dataset] [C12] Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, Gang Hua. *"Ordinal Regression with Multiple Output CNN for Age Estimation,"* in Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2016.

PREPRINT / UNDER-REVIEW PAPERS:

[arXiv] [Github] [X01] Yu Zeng*, Mo Zhou*, Yuan Xue, Vishal M. Patel, *"Securing Deep Generative Models with Universal Adversarial Signature,"*, 2023, Under Review.

PATENTS

[P01] Le Wang, Mo Zhou, Sanping Zhou, Shitao Chen, Jingmin Xin, Nanning Zheng, "A Practical Relative Order Adversarial Attack Method". Chinese Patent No. 202110998691.9.

[P02] Zhenxing Niu, Wei Xue, Mo Zhou, Bo Yuan, Xinbo Gao, Gang Hua, "Age estimation method based on multi-output convolution neural network and ordered regression". Chinese Patent No. 201610273524.7.

ACTIVITIES

- Reviewer of International Conferences
  - IEEE Conf. on Computer Vision and Pattern Recognition (CVPR) 2020 – 2025
  - Annual Conf. on Neural Information Processing Systems (NeurIPS) 2022 – 2025
  - International Conf. on Computer Vision (ICCV) 2021 – 2025
  - European Conf. on Computer Vision (ECCV) 2020 – 2024
  - International Conf. Learning Representations (ICLR) 2022 – 2025
  - International Conf. of Machine Learning (ICML) 2023 – 2024
  - Others, incl.: AAAI, WACV, ACCV, ICPR, etc. 2021 – 2025
- Reviewer of International Journals
  - IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI) 2021 – 2023
  - Springer Journal: International Journal of Computer Vision (IJCV) 2023 – 2025
  - IEEE Trans. on Dependable and Secure Computing (TDSC) 2022
  - Others, incl.: TNNLS, TMM, NeuNet, NeuComp, MVA, CAIS, etc. 2021 – 2024
- Organizer of International Workshop and Competition
  - [Website] Erasing the Invisible: A Stress-Test Challenge for Image Watermarks NeurIPS 2024
  - [Website] 4th Workshop of Adversarial Machine Learning on Computer Vision CVPR 2024
  - [Website] 4th Workshop on Adversarial Robustness In the Real World ICCV 2023
- Volunteer in Free and Open-Source Software Communities
  - Official Developer for Debian GNU/Linux 08/2018 – Current
  - Contributor for Gentoo GNU/Linux 06/2019 – 08/2019
  - Google Summer of Code (GSoC) as Mentor w/ Debian Project 2025
    Project: *Packaging LLM Inference Libraries* (Student: Kohei Sendai)
  - Open Source Promotion Plan (OSPP) w/ Tsinghua University TUNA Association 2020
    Project: *Integrating Data Science Software into Debian* (Best Quality Award)
  - Google Summer of Code (GSoC) as Student w/ Debian Project 2020
    Project: *BLAS/LAPACK Ecosystem Enhancement for Debian*
  - Google Summer of Code (GSoC) as Student w/ Gentoo Foundation 2019
    Project: *BLAS and LAPACK Runtime Switching*

TEACHING

- Deep Learning (`EN.520.638.01.SP25`), Johns Hopkins University Spring 2025
  Teaching Assistant for Prof. Vishal M. Patel

Honors

- Outstanding Reviewer for CVPR 2024 — 2024
- Outstanding Reviewer for ICCV 2021 — 2021
- Xidian University Secondary School Scholarship.+ — 2017-2018
- Interdisciplinary Contest in Modeling (ICM) — 2016
  Meritorious Winner. Advisor: Youlong Yang (Xidian University)

Affiliation

- IEEE Student Member, IEEE — Aug 2021 – Dec 2025

References

Available upon request.